



UJIAN TENGAH SEMESTER PENDEK
 PEMBELAJARAN MENGGUNAKAN TIK
 TAHUN AKADEMIK 2024/2025

Mata Kuliah/SKS	: Lab. Kecerdasan Buatan
Fakultas	: Ilmu Komputer
Program/ Semester	: Sistem Informasi / 5 PSIA-5PSIB
Hari/Tanggal	: XX, XX
Sifat Ujian	: Project
Dosen Penguji	: Tim Dosen

CPMK 1: Students are able to evaluate (C5) AI methods and algorithms by conducting (P2) simulations and complying with (A2) procedures

- Fuzzy Logic-Based Expert System for Adaptive Cruise Control in cars
Modern vehicles are equipped with **Adaptive Cruise Control (ACC)** systems designed to automatically adjust the vehicle’s speed to maintain a safe distance from the car ahead. A fuzzy logic controller can be implemented to make smooth, human-like decisions based on the current traffic situation.

The controller uses the minimum of three inputs variables:

- Input 1:** *Distance (D)* - the distance to the vehicle ahead (in meters).
- Input 2:** *Relative Speed (ΔV)* – the speed difference between our car and the vehicle ahead (in km/h).
- Input 3:** Road Slope (S) — the inclination of the road (in degrees), where positive values represent uphill and negative values represent downhill conditions.
- Output:** *Acceleration Command (A)* – determines whether the vehicle should accelerate, maintain speed, or decelerate.

- Definition of Variables
Define the fuzzy linguistic variables and their corresponding fuzzy sets for each input and output. Provide appropriate **membership functions** and describe their ranges.
- Rule Base Design
Construct at least **nine (9) fuzzy IF–THEN rules** that describe the decision-making behavior of the ACC system. Rules must reflect realistic driving behavior and safe operation.
- Evaluation and Discussion
Interpret the resulting output.
 - Does the system’s decision align with real-world driving logic?
 - Explain how fuzzy logic offers a more flexible and human-like control behavior compared to traditional threshold-based systems.

CPMK 2: Students are able to evaluate (C5) ML applications by conducting (P2) experiments and complying with (A2) procedures

- Linear Regression Design for Vehicle Fuel Efficiency Prediction
Automotive engineers aim to improve **fuel efficiency (measured in km per liter)** by analyzing the relationships between vehicle design parameters and performance factors. You are assigned to develop a **linear regression model** to predict the **fuel efficiency (Y)** of a vehicle based on several influencing variables.

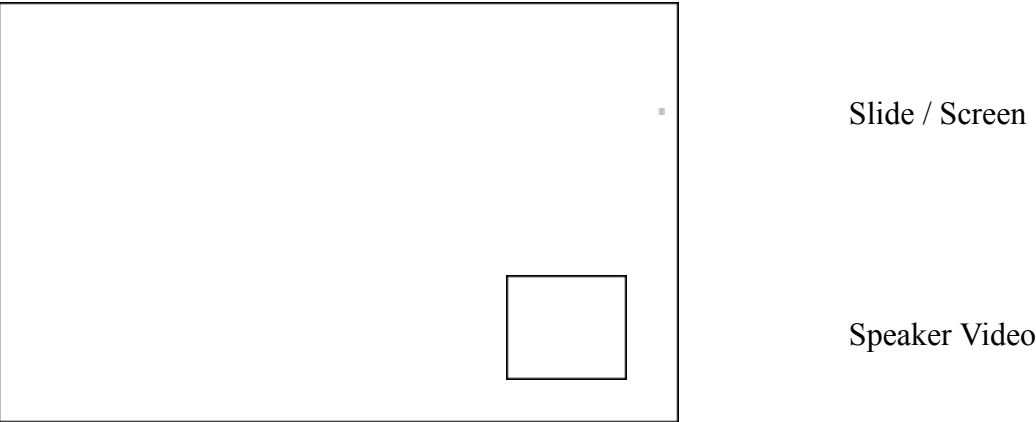
You may collect or download your own dataset from a reliable source such as:

- Kaggle: Vehicle Fuel Efficiency / MPG Dataset
- UCI Machine Learning Repository: Auto MPG Dataset
- or other open datasets with relevant attributes.

- Data Preprocessing
 - Perform **data cleaning** (handle missing values, remove duplicates, and correct data types).
 - Conduct **data transformation**, such as normalization, encoding categorical variables, or scaling numerical values.
- Model Development
 - Split the dataset into training and testing subsets (e.g., 80%–20%).
 - Train a **Linear Regression model** using the selected features.
- Model Evaluation

- Evaluate the model using **performance metrics**, such as:
 - Mean Absolute Error (MAE)
 - Mean Squared Error (MSE)
 - Root Mean Squared Error (RMSE)
 - R^2 Score
 - Discuss the performance results and explain whether the model fits the data well.
- d. Interpret the final result on what factors most influence vehicle fuel efficiency?

Record your presentation in form of video, the video layout is as follows:



- Submit instructions:
1. Both notebooks must be written in **Python** (Jupyter Notebook or Google Colab).
 2. All notebooks must be executable from top to bottom without error.
 3. For **Question 2** (Vehicle Fuel Efficiency Prediction), include the dataset file in the same folder as the notebook.
 4. Compress the folder into a single ZIP file before submission.
 5. If your ZIP File are to large, then upload it on GDrive and submit the link with a .txt fille instead

Assessment Matric

Criteria	Below Expectation	Meet Expectation	Exceed Expectation
Fuzzy Logic-Based Expert System (40)	Implementation is incomplete, lacks defined membership functions or fuzzy rules, and produces invalid or unclear results. (0–20)	The system includes basic fuzzy logic components with correct rule base and produces acceptable results. (21–30)	The system is well-designed with accurate rules, clear reasoning, and meaningful inference results. (31–40)
Vehicle Fuel Efficiency Prediction (40)	Dataset is not properly cleaned or transformed, model training or evaluation is missing or incorrect. (0–25)	Dataset preprocessing, model training, and evaluation are done correctly with basic interpretation of results. (26–40)	The notebook demonstrates strong understanding of regression design with proper data processing, model tuning, and insightful analysis. (41–50)
Report and Team Contribution (10)	Documentation is unclear or incomplete, and team contribution is not specified. (0–4)	Report is structured and includes a team contribution list. (5–7)	Report is comprehensive, well-written, and includes balanced team contribution with reflection. (8–10)
Presentation (10)	Presentation are not cleared and presenter seems reading (0-4)	Presentation are delivered quite well (5-7)	Presentation are Delivered interestingly (8-10)